

Scalability & Future Plan

Date: 29 September 2026	Team ID: SWTID-2026-9692
Project Name: WeatherWise (ai-weatherwise-api)	Maximum Marks: 1 Mark

Scalability & Future Plan

This document outlines how the current WeatherWise project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future [cite: 43]. It also captures the team's roadmap for enhancing and evolving the project beyond its current state [cite: 43].

Current System Limitations

S.No	Limitation	Impact	Priority
1	Deprecated AI Model Endpoint	API requests to gemini-2.5-flash result in 404 errors, causing fallback responses in weather summaries and recommendations.	High
2	Single Node Execution (Localhost)	Backend runs on single-node Express server on port 5001 with local MongoDB instance (127.0.0.1), limiting concurrent user handling.	High
3	Lack of External Weather API Caching	Repeated queries for the same city hit third-party weather services directly without Redis caching, increasing latency and API costs.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single instance Node.js server.	Deploy containerized microservices via Docker/Kubernetes with NGINX load balancer for horizontal scaling.
2	Data Storage	Local standalone MongoDB.	Migrate to MongoDB Atlas Multi-Region Cluster with database sharding and index optimization.
3	Performance	Direct weather API queries & synchronous AI execution.	Implement Redis Caching for weather data (TTL 15 mins) and asynchronous task queues (BullMQ) for heavy AI processing.
4	Security	Basic JWT Authentication without rate limiting.	Add HTTPS/TLS, Express Rate Limiting, API Gateway throttling, and secure secret management (AWS Secrets Manager).

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Upgrade AI model integration to latest Gemini APIs, implement Redis caching, and deploy REST API to AWS/Render.	Q4 2026	Restores full AI accuracy and improves request latency by 60%.
Phase 3	Add multi-city severe weather alert notifications (Email/WebPush) and historical weather trend analytics.	Q1 2027	Increases daily active user engagement and safety utility.
Phase 4	Develop cross-platform Flutter mobile app and personalized location-based weather AI widgets.	Q2 2027	Expands user footprint across mobile platforms.